Q1 - Single Correct

If the roots of the equation $ax^2 + bx + 10 = 0$ are not real and distinct, where $a, b \in R$ and α and β are values of a and b for which $5a + b$ is minimum, then the family of lines $\alpha(4x + 2y + 1) + \beta(x - y - 1) = 0$ are concurrent at

(1) (1,-1)

(2) (1,1)

(3) $\left(-\frac{1}{6}, -\frac{7}{6}\right)$

(4) $\left(\frac{1}{6}, \frac{7}{6}\right)$

Q2 - Single Correct

Let $A(1, 1)$ and $B(3, 2)$ be two points. If C is a point on X -axis such that $AC + BC$ is minimum, then the coordinates are

(1) $\left(\frac{5}{3}, 0\right)$

(2) $\left(\frac{1}{3}, 0\right)$

(3) (3,0)

(4) None of these

Q3 - Single Correct

Let A and B have coordinates (x_1, y_1) and (x_2, y_2), respectively. We define the distance between A and B as $d(A, B) = \max\{|x_2 - x_1|, |y_2 - y_1|\}$. If $d(0, A) = 1$, where 0 is origin, then the locus of A is

(1) a square

(2) pair of parallel lines

(3) a square of area 2 sq units

(4) a square of area 8 sq units

Q4 - Single Correct

m, n are integers with $0 < n < m$. A is the point (m, n) on the cartesian plane. B is the reflection of A in the line $y = x$. C is the reflection of B in the Y -axis, D is the reflection of C in the X -axis and E is the reflection of D in the Y -axis. The area of the pentagon $ABCDE$, is

(1) $2m(m+n)$

(2) $m(m+3n)$

(3) $m(2m+3n)$

(4) None of these

Q5 - Single Correct

The distance between the two parallel lines is 1 unit. A point A is chosen to lie between the lines at a distance d from one of them. ΔABC is equilateral with B on one line and C on the other parallel line. Then, length of the side of the equilateral triangle is

(1) $\frac{2}{3}\sqrt{d^2+d+1}$

(2) $2\sqrt{\frac{d^2-d+1}{3}}$

(3) $2\sqrt{d^2+d+1}$

(4) $\sqrt{d^2-d+1}$

Q6 - Single Correct

Number of values of λ such that $\left(\lambda^2+1, \lambda\right)$ is the mirror image of $(\lambda, \lambda+1)$ w.r.t. $x+2y-1=0$

(1) 0

(2) 1

(3) 2

(4) 3

Q7 - Single Correct

The number of triangles having two vertices as (1,2) and (6,2) and incentre (4,6), is

(1) 2

(2) 1

(3) infinite

(4) 0

Q8 - Single Correct

If AD, BE and CF are the altitudes of ΔABC whose vertex A is the point (-4,5) . The coordinates of the points E and F are (4,1) and (-1,-4) respectively. Then, equation of BC is

(1) $3x-4y-28=0$

(2) $4x+3y+28=0$

(3) $3x-4y+28=0$

(4) $x+2y+7=0$

Q9 - Single Correct

The vertices of a triangle are $A(m, n), B(12,19)$ and $C(23,20)$, where m and n are integers. If its area is 70 sq units and the slope of the median through A is -5, then the last digit of $(m+n)$ is

(1) 3

(2) 5

(3) 7

(4) 8

Q10 - Single Correct

If the centroid and circumcentre of a triangle are (3,3) and (6,2) respectively, then the orthocentre is

(1) (-3,5)

(2) (-3,1)

(3) (3,-1)

(4) (9,5)

Q11 - Single Correct

A and B are fixed points. The vertex C of ΔABC moves such that $\cot A+\cot B=$ constant. The locus of C is a straight line is

(1) perpendicular to AB

(2) parallel to AB

(3) inclined at an angle $(A-B)$ to AB

(4) None of these

Q12 - Single Correct

The number of integral points (integral point means both the coordinates should be integer) exactly in the interior of the triangle with the vertices (0,0),(0,21) and (21,0) is

(1) 133

(2) 190

(3) 233

(4) 105

Q13 - Single Correct

The three lines whose combined equation is $y^3-4x^2y=0$ form a triangle which is

(1) isosceles

(2) equilateral

(3) right angled

(4) None of these

Q14 - Single Correct

If the distance of a point (x_1, y_1) from each of two straight lines, which pass through the origin of coordinates, is δ. Then, the two lines are given by

(1) $(x_1y-xy_1)^2=\delta^2(x+y)^2$

(2) $\left(x_1 y+x y_1\right)^2=\delta^2\left(x^2+y^2\right)$

(3) $\left(x_1 y-x y_1\right)^2=\delta^2\left(x^2-y^2\right)$

(4) None of these

Q15 - Single Correct

The product of perpendiculars drawn from the origin to the lines represented by the equation $a x^2+2 h x y+b y^2+2 g x+2 f y+c=0$, will be

(1) $\frac{a b}{\sqrt{a^2-b^2+4 h^2}}$

(2) $\frac{b c}{\sqrt{b^2-c^2+4 h^2}}$

(3) $\frac{c}{\sqrt{(a-b)^2+4 h^2}}$

(4) None of these

Q16 - Multiple Correct

The equations of the sides $A B, B C$ and $C A$ of a $\Delta A B C$ are $2 x+y=0, x+p y=q$ and $x-y=3$, respectively and if the centroid is $(2,3)$, then

(1) $p+q=74$

(2) equation of $B C$ is $x+15 y+59=0$

(3) equation of median through A is $5 x-y-7=0$

(4) equation of altitude through A is $11 x-y-13=0$

Q17 - Multiple Correct

Two sides of a rhombus OABC (lying entirely in first quadrant or fourth quadrant) of area equal to 2 sq units, are $y=\frac{x}{\sqrt{3}}$ and $y=\sqrt{3} x$. Then, possible coordinates of B is/are (O being origin)

(1) $(1+\sqrt{3}, 1+\sqrt{3})$

(2) $(-1-\sqrt{3},-1-\sqrt{3})$

(3) $(\sqrt{3}-1, \sqrt{3}-1)$

(4) None of these

Q18 - Multiple Correct

If the lines $ax + by + c = 0, bx + cy + a = 0$ and $cx + ay + b = 0$ are concurrent $(a + b + c \neq 0)$. Then,

(1) $a^3 + b^3 + c^3 - 3abc = 0$

(2) $a = b$

(3) $a = b = c$

(4) $a^2 + b^2 + c^2 - ab - bc - ca = 0$

Q19 - Multiple Correct

If the point (α, α^2) lies inside the triangle formed by the lines $2x + 3y - 1 = 0$, $x + 2y - 3 = 0$ and $5x - 6y - 1 = 0$, then α can be in the interval

(1) $\left(-\infty, -\frac{3}{2}\right)$

(2) $\left(-\frac{3}{2}, -1\right)$

(3) $(1, \infty)$

(4) $\left(\frac{1}{2}, 1\right)$

Q20 - Multiple Correct

Two lines from the family of lines $(1 + 2\lambda)x - (1 + \lambda)y + 1 = 0$ and the line $x + y = 5$ form an equilateral triangle. The equation of the two lines can be

(1) $y - 2 = (2 + \sqrt{3})(x - 1)$

(2) $y - 2 = (2 - \sqrt{3})(x - 1)$

(3) $y - 2 = (3 - \sqrt{2})(x - 1)$

(4) $y - 2 = (3 - \sqrt{2})(x - 2)$

Q21 - Multiple Correct

The diagonals of rhombus $ABCD$ intersect at point (1,2) and its sides are parallel to the lines $x - \sqrt{3}y + 2\sqrt{3} = 0$ and $\sqrt{3}x - y + 3 = 0$. If the vertex A lies on X -axis, then possible coordinates of vertex C are

(1) (-1,4)

(2) (-1,-4)

(3) (3,4)

(4) None of these

Q22 - Multiple Correct

The area of a triangle is 5 sq units. Two of its vertices are (2,1) and (3,-2) the third vertex lies on $y = x + 3$. The coordinates of the third vertex can be

(1) $\left(-\frac{3}{2}, \frac{3}{2}\right)$
(2) $\left(\frac{3}{4}, -\frac{3}{2}\right)$
(3) $\left(\frac{7}{2}, \frac{13}{2}\right)$
(4) $\left(-\frac{1}{4}, \frac{11}{4}\right)$

Q23 - Multiple Correct

The straight lines $x + y = 0, 3x + y - 4 = 0$ and $x + 3y - 4 = 0$ form a triangle which is

(1) isosceles

(2) right angled

(3) obtuse angled

(4) equilateral

Q24 - Multiple Correct

The x -coordinates of the vertices of a square of unit area are the roots of the equation $x^2 - 3|x| + 2 = 0$ and the y -coordinates of the vertices are the roots of the equation $y^2 - 3y + 2 = 0$. Then, the possible vertices of the square is/are

(1) (1,1),(2,1),(2,2),(1,2)

(2) (-1,1),(-2,1),(-2,2),(-1,2)

(3) (2,1),(1,-1),(1,2),(2,2)

(4) (-2,1),(-1,-1),(-1,2),(-2,2)

Q25 - Multiple Correct

Let P be (5,3) and a point R on $y=x$ and Q on X -axis is such that $PQ+QR+RP$ is minimum. Then, coordinates of Q are

(1) $\left(\frac{17}{4}, 0\right)$

(2) (17,0)

(3) $\left(\frac{17}{2}, 0\right)$

(4) None of these

Q26 - Paragraph

Passage I (For Question 26, 27)

ABC is a triangle right angled at A with vertices A, B, C in the anti-clockwise sense in that order. $A(1,2), B(-3,1)$ and vertex C lies on the X -axis. $BCEF$ is a square with vertices B, C, E, F in the clockwise sense in that order. ACD is an equilateral triangle with vertices A, C, D in the anti-clockwise sense in that order.

Slope of AF is

(1) $7/10$

(2) $7/9$

(3) $9/10$

(4) $11/10$

Q27 - Paragraph

The abscissa of centroid of $\triangle BCE$ is

(1) -1

(2) $-1/2$

(3) $-1/3$

(4) $-2/3$

Q28 - Paragraph

Passage II (For Question 28, 29)

Consider a family of lines $(4a+3)x-(a+1)y-(2a+1)=0$, where $a \in R$.

The locus of the foot of the perpendicular from the origin on each member of this family, is

(1) $(2x-1)^2+4(y+1)^2=5$

(2) $(2x-1)^2+(y+1)^2=5$

(3) $(2x+1)^2+4(y-1)^2=5$

(4) $(2x-1)^2+4(y-1)^2=5$

Q29 - Paragraph

A member of this family with positive gradient making an angle of $\frac{\pi}{4}$ with the line $3x-4y=2$, is

(1) $7x-y-5=0$

(2) $4x-3y+2=0$

(3) $x+7y=15$

(4) $5x-3y-4=0$

Q30 - Paragraph

Passage III (For Question 30, 31)

A variable line L is drawn through $O(0,0)$ to meet the line $L_1: y-x-10=0$ $L_2: y-x-20=0$ at the points A and B, respectively. A point P is taken on L

Locus of P, if $\frac{2}{\mathrm{OP}}=\frac{1}{\mathrm{OA}}+\frac{1}{\mathrm{OB}}$, is

(1) $3x+3y=40$

(2) $3x+3y+40=0$

(3) $3x-3y=40$

(4) $3x-3y+40=0$

Q31 - Paragraph

Locus of P, if $OP^2=OA\cdot OB$, is

(1) $(y-x)^2=100$

(2) $(y-x)^2=50$

(3) $(y-x)^2=200$

(4) $(x-y)^2=20$

Q32 - Paragraph

Passage IV (For Question 32, 33)

For points $P\left(x_1,y_1\right)$ and $Q\left(x_2,y_2\right)$ of the coordinate plane, a new distance $d(P,Q)$ is defined by d $(P,Q)=\left|x_1-x_2\right|+\left|y_1-y_2\right|$. Let $O(0,0), A(1,2), B(2,3)$ and $C(4,3)$ are four points on XY -plane.

Let $R(x,y)$, such that R is equidistant from the points O and A with respect to new distance and if $0\leq x<1, 0\leq y<2$, then R lies on a line segment whose equation is

(1) $x+y=3$

(2) $x+2y=3$

(3) $2x+y=3$

(4) $2x+2y=3$

Q33 - Paragraph

Let $S(x,y)$, such that S is equidistant from points O and B with respect to new distance and if $x\geq 2$ and $O\leq y<3$, then locus of S is

(1) a line segment of finite length

(2) a line of the infinite length

(3) a ray of finite length

(4) a ray of infinite length

Q34 - Paragraph

Passage V (For Question 34, 35)

A straight line passing through $O(0,0)$ cuts the lines $x=\alpha, y=\beta$ and $x+y=8$ at A, B and C respectively such that $OA \cdot OB \cdot OC=48\sqrt{2}$ and $f(\alpha, \beta) \leq 0$, where
$f(x, y)=\left|\frac{y}{x}-\frac{3}{2}\right|+(3x-2y)^6+\sqrt{ex+2y-2e-6}$

The point of intersection of lines $x=\alpha$ and $y=\beta$, is

(1) (3,2)

(2) (2,3)

(3) (3,4)

(4) (5,3)

Q35 - Paragraph

Equation of line OA, is

(1) $x-y=0$

(2) $x-\sqrt{2}y=0$

(3) $\sqrt{2}x-y=0$

(4) $2x-y=0$

Q36 - Paragraph

Passage VI (For Question 36, 37)

Diagonal AC of rhombus $ABCD$ is a member of both the family of lines $L_1+\lambda L_2=0$ and $L_3+\mu L_4=0$ and vertex B of rhombus is $(3,2)$. Suppose,

$L_1 : x + y - 1 = 0$
$L_2 : 2x + 3y - 2 = 0$
$L_3 : x - y + 2 = 0$
$L_4 : 2x - 3y + 5 = 0$

The equation of diagonal AC, is

(1) $2x + y + 1 = 0$

(2) $x + 2y + 3 = 0$

(3) $x + 2y - 1 = 0$

(4) $2x + y - 7 = 0$

Q37 - Paragraph

The equation of diagonal BD, is

(1) $2x + y + 1 = 0$

(2) $2x - y - 4 = 0$

(3) $x - 2y + 7 = 0$

(4) $x + 2y - 1 = 0$

Q38 - Integer Type

Number of lines that can be drawn through the point (4,-5) so that its distance from (-2,3) will be equal to 12, is equal to

Q39 - Integer Type

Let slope of sides AB and BC of a ΔABC are roots of the equation $x^2 - 4x + 1 = 0$, (ΔABC lying in the first quadrant and B is origin) and in radius of triangle is 1 unit. If coordinates of its incentre are (a, b). Then, find $\left(a^2 + b^4\right)$.

Q40 - Integer Type

The ΔABC, right angled at C has median AD, BE and CF. AD lies along the line $y=x+3$ BE lies along the line $y=2x+4$. If the length of the hypotenuse is 60, the area of ΔABC is abc sq units, then $(a+b+c)$ is

Q41 - Integer Type

Let $x \tan \alpha + y \sin \alpha = \alpha, \alpha x \operatorname{cosec} \alpha + y \cos \alpha = 1$. Let P be the point of intersection of the lines in the limiting position when $\alpha \to 0$, if the point P is (h, k), then $|h+k|$ is

Q42 - Integer Type

A ΔABC having vertices of $A(5,1), B(-1,-7)$ and $C(1,4)$, respectively. L be the line mirror passing through C and parallel to AB and a light ray emanating from point A goes along the direction of internal bisector of the $\angle A$, which meets the mirror and BC at E and D respectively. If sum of the areas of ΔACE and ΔABC is $(10+a)$ sq unit. Then, a is equal to

Q43 - Integer Type

If the line $y=\sqrt{3}x$ cuts the curve $x^4+ax^2y+bxy+cx+dy+6=0$ at A, B, C and D, then the value of $\frac{1}{12}\mathrm{OA} \cdot \mathrm{OB} \cdot \mathrm{OC} \cdot \mathrm{OD}$ (where, O is the origin) is equal to

Q44 - Integer Type

Two mutually perpendicular straight lines through the origin form an isosceles triangle with the line $2x+y=5$. Then, the area of the triangle, is

Q45 - Integer Type

A ΔABC is formed by three lines $x+y+2=0, x-2y+5=0$ and $7x+y-10=0$ P is a point inside the ΔABC such that areas of $\Delta PAB, \Delta PBC$ and ΔPCA are equal. If the coordinates of the point P are (a, b) and the area of the ΔABC is δ, then $(a+b+\delta-10)$, is

Q46 - Integer Type

A triangle is formed by the line $x+y=0, x-y=0$ and $px+qy=1$. If $p^2+q^2=1$, $\left(p^2-q^2\neq 0\right)$ and locus of its circumcentre is $\left(x^2+y^2\right)=k\left(x^2-y^2\right)^2$, then $(k-1)$ is equal to

Q47 - Integer Type

$A_1, A_2, A_3, \ldots, A_n$ are points on the line $y=x$ lying in the positive quadrant such that $OA_n=n\cdot OA_{n-1}$, "O" being the origin. If $OA_1=1$ and the coordinates of A_n are $[2520\sqrt{2}, 2520\sqrt{2}]$, then the value of n is

Q48 - Integer Type

The combined equation of three sides of a triangle is $\left(x^2-y^2\right)(2x+3y-6)=0. f(-2,a)$ is the interior point of the triangle, then the integer value of a is

Q49 - Integer Type

The value of λ for which the lines joining the point of intersection of curves C_1 and C_2 to the origin are equally inclined to the axis of $X\cdot C_1: \lambda x^2+3y^2-2\lambda xy+9x=0$ and $C_2: 3x^2-4y^2+8xy-3x=0$, then λ is equal to

Q50 - Integer Type

The equation of the locus of foot of perpendiculars drawn from the origin to the line passing through a fixed point (a,b) is $x^2+y^2-ax-by=\lambda$, then λ is equal to

Answer Key

Q1 (3)	**Q2** (1)	**Q3** (1)	**Q4** (2)
Q5 (2)	**Q6** (1)	**Q7** (4)	**Q8** (1)
Q9 (3)	**Q10** (1)	**Q11** (2)	**Q12** (2)
Q13 (4)	**Q14** (4)	**Q15** (3)	**Q16** (1,3,4)
Q17 (1,2)	**Q18** (1,2,3,4)	**Q19** (2,4)	**Q20** (1,2)
Q21 (1,3)	**Q22** (1,3)	**Q23** (1,3)	**Q24** (1,2)
Q25 (1)	**Q26** (4)	**Q27** (3)	**Q28** (4)
Q29 (1)	**Q30** (4)	**Q31** (3)	**Q32** (4)
Q33 (4)	**Q34** (2)	**Q35** (1)	**Q36** (3)
Q37 (2)	**Q38** (0)	**Q39** (6)	**Q40** (4)
Q41 (1)	**Q42** (9)	**Q43** (8)	**Q44** (5)
Q45 (5)	**Q46** (0)	**Q47** (7)	**Q48** (3)
Q49 (12)	**Q50** (0)		